

Large-Scale Deployment of LPO in AI Clusters

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Abstract We present the large-scale deployment of 400G DR4 LPO modules in AI clusters, reporting statistical data on system conditions, link performance, and reliability from over 25.8 million device-hours of field operation, proving the long-term reliability of LPO technology. ©2026 The Author(s)

Introduction

As artificial intelligence (AI) and high-performance computing (HPC) clusters evolve towards ultra-large scales and higher compute densities [1], both scale-out and scale-up networks face stringent requirements for power efficiency, latency, reliability [2], cost, and deployment efficiency in optical interconnects. Traditional digital signal processor (DSP) based fully retimed pluggable optics (FRO) are increasingly struggling to meet these demands. Linear-drive pluggable optics (LPO) technology addresses this by eliminating power-hungry and costly DSP chips, utilizing linear transimpedance amplifiers (TIAs) and drivers to directly drive optical components. This architecture significantly reduces power, latency, and cost while retaining the operational flexibility of pluggable optics. More critically, compared to alternative solutions such as co-packaged optics (CPO), the LPO ecosystem is more mature, enabling rapid large-scale deployment. Consequently, LPO is a pivotal technology for next-generation AI optical interconnects.

This paper presents the large-scale deployment of 400G DR4 LPO modules in a production AI environment. We detail the deployment environment and system conditions, present comprehensive link performance metrics, and analyze long-term reliability statistics. Our field data validate the maturity and robustness of LPO technology for mass deployment, providing critical empirical benchmarks for the industry.

System Overview

Both the deployed LPO and the benchmark FRO are deployed in AI scale-out fabrics [3], interconnecting Top-of-rack (ToR) switches with aggregation (Agg) switches, and Agg switches with core switches. Notably, the ToR and Agg switches share identical hardware. The switch is built around the Broadcom Tomahawk 5 (TH5) 51.2T switching ASIC, housed in a 4U chassis with 128 front-panel 400G QSFP112 ports. As illustrated in Fig. 1(a), the hardware architecture features two distinct signal paths: the upper 88 ports in-

terface directly with the TH5 ASIC on a single media access control (MAC) board, whereas the lower 40 ports reside on an independent line card, connecting vertically to the MAC board via mezzanine connectors, which introduce additional signal impairments. Fig. 1(b) depicts the insertion loss distribution across all channels, revealing that the maximum loss approaches the 16 dB limit specified by the CEI-112G-VSR standard. Crucially, this switch is not custom-designed for LPO applications; instead, it is a generic, low-cost, and PHY-less switch. LPO are based on silicon photonics technology sourced from three distinct vendors. The key components, including drivers, TIAs, and photonic integrated circuits (PICs), are supplied by various manufacturers. This validates multi-vendor LPO interoperability. Regarding module settings, the continuous time linear equalizer (CTLE) in the driver of all LPO is fixed at 0 dB, offloading all channel compensation tasks to the TH5 SerDes receiver equalizer. With respect to interconnect distances, Fig. 1(c) illustrates the length distribution of multi-fiber push-on (MPO) cables used for point-to-point connections, with the longest link reaching 250 meters.

Performance Characterization

Prior to mass deployment, we conducted a comprehensive characterization of LPO performance on TH5 switches in the laboratory. As shown in Fig. 2(a), optical loopback tests were performed on all switch ports and lanes at 106.25 Gb/s PAM4 using the PRBS31Q pattern. The results demonstrate that the bit error rate (BER) for all links is over 2 orders of magnitude below the 2.4×10^{-4} pre-FEC threshold required by the KP4 forward error correction (FEC), confirming robust baseline link performance, notably, due to the impact of additional signal impairments, the BER on the line card is an order of magnitude higher than that on the MAC board. Fig. 2(b) illustrates the transmitter and dispersion eye closure quaternary (TDECQ) values for LPO on typical TH5 switch ports, along with their correlation to BER. The re-

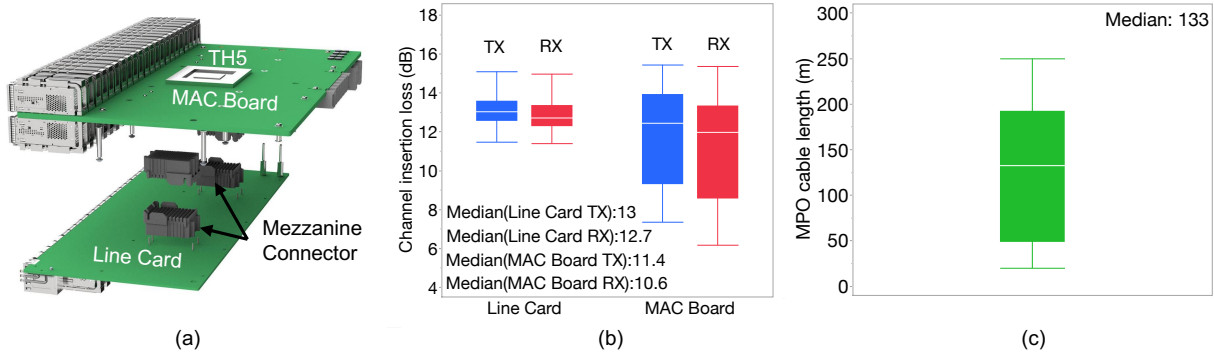


Fig. 1: (a) TH5 switch hardware architecture. (b) Distribution of switch channel insertion loss. (c) Interconnect reach distribution for deployed modules.

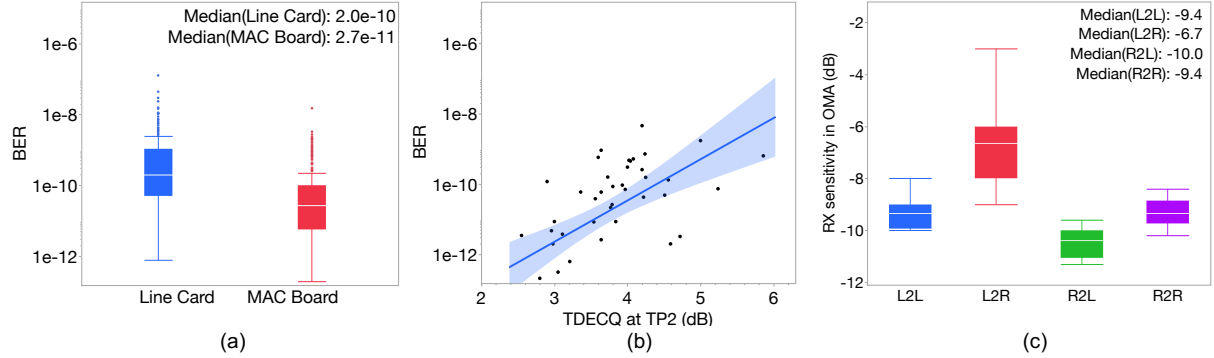


Fig. 2: (a) Comparison of LPO BER across two distinct switch signal paths. (b) Correlation between LPO TDECQ and BER. (c) Interoperability characterization via receiver sensitivity measurements.

sults reveal a weak correlation between TDECQ and BER. In fact, although the TDECQ of certain ports exceeds the 3.5 dB upper limit specified by the IEEE 802.3bs 400GBASE-DR4 standard, the links still exhibit excellent BER performance. Fig. 2(c) details the interoperability characterization of LPO and FRO on TH5 switches, conducted by receiver sensitivity measurements across a full matrix of cross-combinations: L2L (linear Tx to linear Rx), L2R (linear Tx to retimed Rx), R2L (retimed Tx to linear Rx) and R2R (retimed Tx to retimed Rx). Characterization results reveal that the L2R scenario exhibits significant performance degradation, with certain retimed receivers failing to meet the -3.9 dBm optical modulation amplitude (OMA) sensitivity required by the DR4 standard, thereby violating the link budget. To mitigate interoperability risks and performance uncertainties arising from mixed deployments, LPO were deployed exclusively in LPO-to-LPO links, while FRO were restricted to FRO-to-FRO connections.

Field Deployment Results

We conducted a statistical analysis and comparative study on long-term field operational data of LPO and FRO deployed within the same data centers. Crucially, both types of modules were sourced from the same three manufacturers and fabricated using identical production processes, ensuring a fair comparison. As shown in Figs. 3(a) and (b), LPO exhibit superior power efficiency and thermal performance. The median power consumption of LPO drops to 4.2 W, representing

a reduction of 48% compared to the 8.1 W of FRO. Furthermore, the median case temperature of the LPO is only 35.1 °C, which is 10.4 °C lower than that of its FRO counterparts. Consequently, the reduced operating temperature significantly improves laser reliability and lifetime [2][4]. Such an enhancement is critically valuable for ensuring the robustness of AI networks. Fig. 3(c) illustrates that the received optical power distribution for LPO links is comparable to that of FRO links. Notably, while FRO exhibits a higher absolute count of outliers, this is attributed to their larger deployment volume; statistical analysis confirms that the outlier ratios are equivalent for both links. The outliers observed in both link types are primarily caused by dust contamination on fiber connector end-faces. Fig. 3(d) highlights the substantial latency advantage of LPO, a critical attribute for AI networks. While FRO from three distinct DSP vendors exhibit latencies exceeding 100 ns, LPO achieve latencies below 5 ns. Fig. 3(e) demonstrates the long-term link stability of LPO by monitoring the TH5 receiver signal-to-noise ratio (SNR) over an 80-day period, during which two typical LPO links exhibited SNR fluctuations of less than 0.3 dB. Fig. 3(f) illustrates the distribution of cumulative maximum non-zero KP4 FEC bin since link establishment. The data show that the majority of maximum non-zero FEC bin values fall below 10, indicating sufficient performance margin for robust operation. Notably, LPO links on the line card exhibit a slightly higher maximum non-zero FEC bin compared to other link

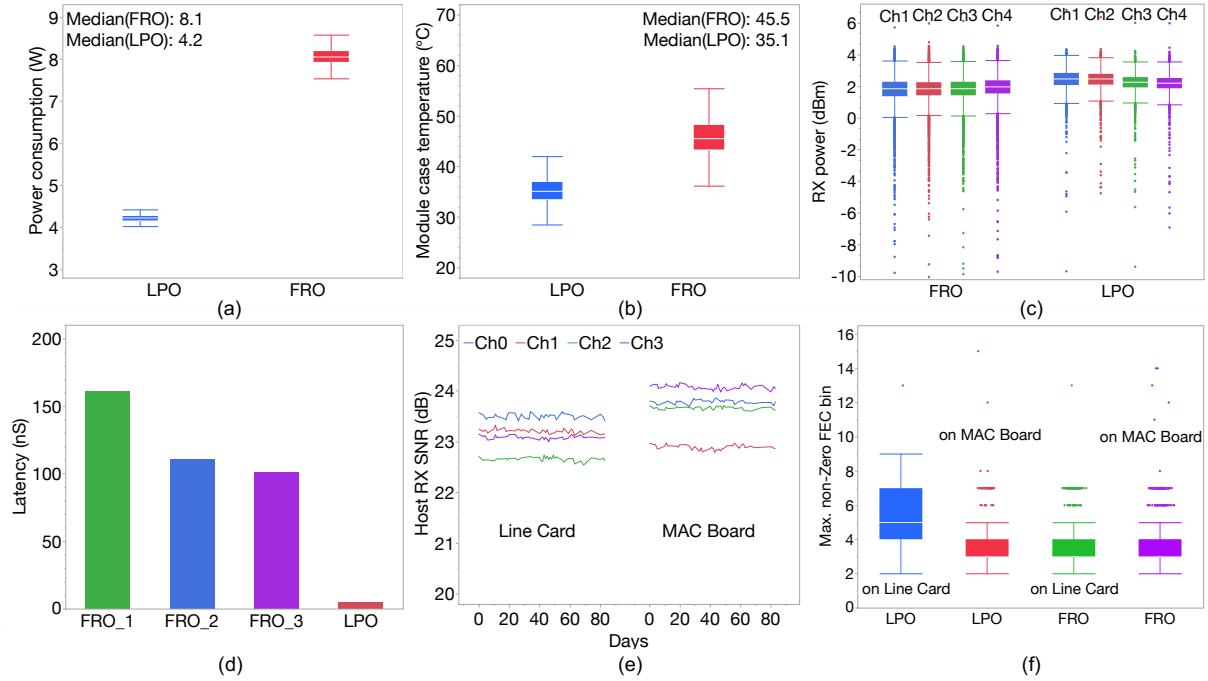


Fig. 3: Statistical distributions of module parameters: (a) power consumption; (b) case temperature; (c) average received optical power; and (d) transmission latency. Also shown are: (e) TH5 receiver SNR stability and (f) distribution of maximum non-zero FEC symbol bins at TH5 receiver.

Tab. 1: Long-term reliability statistics and comparative analysis of LPO versus FRO.

	Device Number	Total Device Hours	RMA Number	RMA In Time	MTB RMA (Years)	Link Flapping Times	Link Flapping In Time	MTB Flapping (Years)	Daily Flapping Rate
LPO	13234	2.58E+07	7	272	420	178	6910	17	0.017%
FRO	220352	8.35E+08	209	250	456	13746	16466	7	0.040%

types, a trend attributed to the impact of the aforementioned additional signal impairments on the line card. Additionally, a small number of outliers observed across all types of links were identified through root cause analysis as being caused by fiber end-face contamination, and these links returned to normal operation after cleaning.

Regarding long-term reliability, as summarized in Table 1, the return merchandise authorization (RMA) in-time data for LPO is 272, which is comparable to that of FRO. When converted to mean time between (MTB RMA), both technologies exceeded 400 years. Root cause analysis of the 7 LPO RMA cases reveals failure modes including electrostatic discharge (ESD) damage to drivers/TIAs, physical damage to fiber end-faces, and contamination. These represent typical hardware failure modes for pluggable optics [5], with a distribution consistent with that of FRO. Notably, monitoring of link flapping data demonstrates the superior link stability of LPO. Link flapping refers to intermittent link down/up events triggered by burst errors; while such events typically do not necessitate hardware replacement, they can significantly impact service stability. Statistical analysis reveals that the link flapping rate for LPO is substantially lower than that of FRO. Specifically, the link flapping in-time

(LFIT) metric is 6,910 for LPO compared to 16,466 for FRO, representing a 58% reduction. This corresponds to a daily flapping rate of 0.017% for LPO versus 0.04% for FRO. We attribute this performance gap primarily to the fact that FRO incorporates DSP chips from multiple vendors, which can introduce interoperability challenges between different DSP implementations.

Conclusions

This paper presents the large-scale deployment of LPO in AI clusters. Based on long-term field monitoring data from 13,234 deployed 400G DR4 LPO, accumulating over 25.8 million device-hours, we demonstrate that eliminating the DSP significantly reduces power consumption and operating temperature while maintaining superior link performance. Statistical analysis reveals that LPO exhibits enhanced link stability compared to traditional FRO. Our findings validate that LPO technology has matured for mass deployment under specific system conditions, particularly in closed and controlled optical interconnect fabrics such as AI scale-up networks. Furthermore, this work paves the way for the evolution of next-generation linear-drive optical interconnect technologies for AI infrastructure.

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